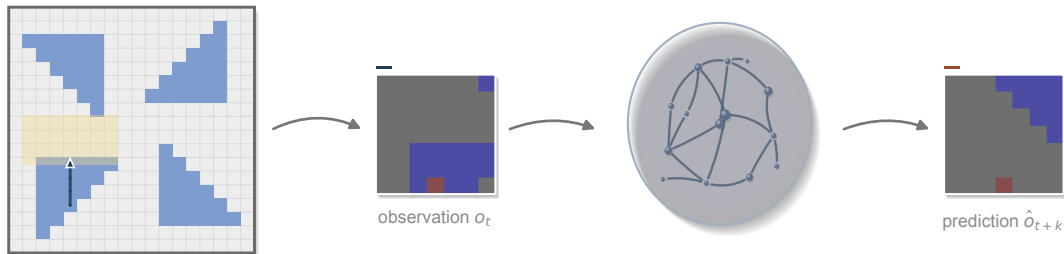


a

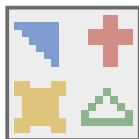


$$a_t = [\text{speed}, \text{onehot}(h)]$$



b

C_1 -- no symmetry



4 distinct landmarks

C_2 -- 180° rotation



2 landmarks, $\times 2$ copies each

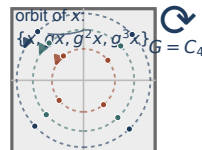
C_4 -- 90° rotation



1 landmark, $\times 4$ copies

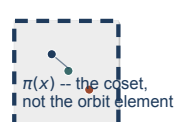
d

state space X



(every point has 4 physically distinct images under G)

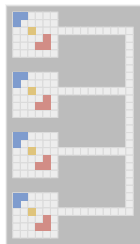
quotient X/G



(one point per orbit -- the fold)

c

Spiers et al. 2015

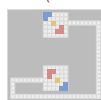


Grievens et al. 2016

parallel (translation)



radial (rotation)



same compass reading
 $\Rightarrow$ **code folds** $\Rightarrow$ **repeats**

different compass reading
 $\Rightarrow$ **code lifts** $\Rightarrow$ **remaps**